2025 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI) | 979-8-3315-2169-1/25/$31.00 ©2025 IEEE | DOI: 10.1109/IATMSI64286.2025.10984687

Machine Learning Models for Predicting Olympic Medal Outcomes

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***Abstract*—In this paper, the prediction of Olympic medal winners has been explored using various machine learning models, utilizing a dataset spanning 128 years of Olympic history from 1896 to 2024. Thirteen Machine Learning models - Logistic Regression, Polynomial Logistic Regression, XGBoost, Random Forest, KNN, Naive Bayes, LightGBM, AdaBoost, Decision Tree, Extra Trees, Gradient Boosting, Neural Network, and SVM were used and evaluated. The dataset was preprocessed, and models were trained and tested to predict future medal outcomes. Superior performance metrics were demonstrated by ensemble models such as XGBoost, LightGBM, and Gradient Boosting with accuracy rates of 83%, 84%, and 84% respectively, and notable AUC values. Conversely, lower performances were exhibited by simpler models like Logistic Regression, Polynomial Logistic Regression, and SVM. Discrepancies in prediction graphs suggested potential issues in model training or dataset encoding. Future research will focus on integrating additional features and more sophisticated techniques to enhance model performance and prediction accuracy. The findings are anticipated to contribute significantly to the field of sports analytics and assist national Olympic committees in strategic planning and resource allocation for future Olympic Games.**

***Keywords*— Machine Learning, Olympic games, medal prediction, Ensemble models, Hyperparameter tuning.**

I. INTRODUCTION

The prediction of Olympic medal winners has intrigued researchers due to its complexity and the valuable insights it offers into sports performance and national athletic programs. Various machine learning models or algorithms have been used to forecast medal counts, leveraging historical data to identify patterns and trends [1], [2], [3]. It has been demonstrated that different boosting algorithms effectively predict Olympic medals, showcasing the potential of machine learning in this domain [1]. Additionally, comparative analyses of machine learning algorithms have provided valuable insights into their predictive capabilities for Olympic outcomes [2]. The motivation for this study arose from the desire to enhance the accuracy of medal predictions by utilizing a comprehensive dataset spanning 128 years of Olympic history from 1896 to 2024, thereby contributing to the field of sports analytics and aiding in the strategic planning of national Olympic committees.

The primary objective of this paper was to evaluate and compare various machine learning models to predict which countries of athletes have a higher probability of winning medals in future Olympic Games. The dataset encompassed 128 years of Olympic history from 1896 to 2024, including athlete demographics, event details, and medal outcomes. 13 Machine Learning models - Logistic Regression, Polynomial Logistic Regression, XGBoost, Random Forest, KNN, Naive Bayes, LightGBM, AdaBoost, Decision Tree, Extra Tree, Gradient Boosting, Neural Network and SVM were used, and they were trained and tested using the dataset as discussed in the Methodology section of this paper. The performance of these models was validated to determine their efficiency. The predicted and actual results of each of these models were compared via data visualization graphs as illustrated in the Results & Discussion section. This study sought to provide a robust framework for predicting Olympic outcomes, thereby offering valuable insights for stakeholders in the sports industry.

II. LITERATURE REVIEW

Different boosting algorithms for predicting Olympic medal outcomes were explored by Sagala and Ibrahim [1], where CatBoost, XGBoost, and LightGBM were compared using historical Olympic data. XGBoost was found to be the most accurate and efficient, highlighting the potential of machine learning in sports analytics to improve prediction accuracy. Similarly, Badoni *et al.* [2] evaluated Random Forest, Decision Trees, and SVMs for medal prediction, with Random Forest demonstrating superior accuracy and robustness. These studies suggested that if machine learning models are tuned appropriately, they could significantly assist national sports committees in strategic planning and resource allocation for future Olympic Games.

A comprehensive analysis of sports data using machine learning techniques was conducted by S. N *et al.* [4] to predict outcomes across various sports, where algorithms such as Gaussian, KNN, Random Forest, and Decision Trees were used. Decision Trees outperformed others in Cricket, while Logistic Regression excelled in Football and Basketball. Similarly, in the paper by Gaur *et al.* [5], the Player Winning Probability Model (PWPM) using Random Forest was proposed, which achieved 70.22% accuracy in predicting Olympic medal outcomes. Data collection, cleaning, and analysis were emphasized, showcasing the potential of machine learning models in sports analytics to enhance performance predictions and strategic planning.

A socio-economic machine learning model was developed by Schlembach *et al.* [6] to forecast Olympic medal distribution during the COVID-19 pandemic using a two-staged Random Forest algorithm, which outperformed traditional forecasts from 2008 to 2016. Socio-economic indicators were incorporated, and predictions for Tokyo 2020 suggested the United States leading, followed by China and Great Britain. Similarly, Shailaja [7] used predictive analytics to evaluate the Olympic performance of India by analyzing GDP, Gini index, population, and other factors. Machine learning models identified economic and social indicators as significant influences on athlete performance, providing insights for improvement and highlighting the value of data-driven decision-making.

Various machine-learning techniques were applied by M. M *et al.* [8] to analyze and predict Olympic results using a dataset spanning over a century of Olympic history. Data preprocessing, including cleaning and feature engineering, was conducted before employing regression, classification, and clustering algorithms to predict medal winners, classify athlete performance, and group sports disciplines. Hidden patterns and trends were identified, demonstrating the potential of machine learning in forecasting medal counts and analyzing Olympic participation. Similarly, Scelles *et al.* [9] utilized econometric models, including the Hurdle and Tobit models, to analyze national medal totals from 1992 to 2016. Socioeconomic, political, and regional variables were incorporated, revealing significant impacts on medal counts. Negative effects were observed in most regions except North America, Oceania, and Western Europe. Consistency with previous predictions for the 2020 Olympics validated the approach, providing a comprehensive understanding of Olympic success determinants. Guo et al. [10] analyzed the challenges of the Hangzhou Asian Games using interdisciplinary methods, correlating elite sports data with COVID-19 statistics. Findings indicated an upward trend in gold medal proportions and medal shares for Asian countries, while the impact of the pandemic on the Games was predicted to be less severe than the global average, highlighting their competitive importance.

III. METHODOLOGY

The dataset used in this study was obtained from Kaggle, which contained data on 128 years of Olympic history from 1896 to 2024, including athlete demographics, event details, and medal outcomes. The dataset contained the following columns: Gender, Age, Height, Weight, National Olympic Committee (NOC) code, Year, Season, Host City, Event, and Medal type (Gold, Silver, Bronze, or NaN), and Medal.

Two CSV files, one covering 1896 to 2022 from GitHub [11] and another for 2024 from Kaggle [12], were merged into a single dataset [13], [14]. The data was sorted in ascending order by year, and any missing values were addressed. Specific cleaning steps included removing leading/trailing spaces in the Medal column and converting the entries to title case. Duplicates were identified and handled based on NOC, Year, Season, Host City, Medal, and Events. One-hot encoding was applied to the Medal column to convert categorical data into a numerical format for model compatibility.

For data visualization, horizontal bar graphs were created to visualize the relationship between various features (e.g., Gender, Age, Height, Weight, NOC, Year, Season, Host City, and Event) and Medal counts.

From Fig. 1(a), it can be seen that the USA, the Soviet Union, and Germany are the top 3 countries that have won the highest amount of medals from 1896 to 2024. In Fig. 1(b), it is observed that participants at the ages of 24, 23, and 25 have won the most medals in those 128 years.

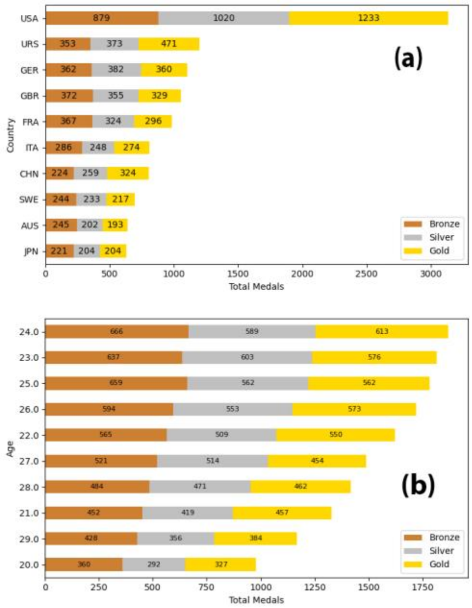

Fig. 1. Top 10 medal-winning (a) countries and (b) age groups for the period 1896–2024 [15].

After that, a correlation heatmap was generated to examine the relationships between numerical variables within the dataset. This analysis helped identify any strong correlations that might influence the model training process, ensuring that multi-collinearity was addressed. In Fig. 2, high positive correlations (0.76) were observed between height and weight and vice versa, moderate correlations (0.18) were seen between age and weight and vice versa, and low or negative correlations (-0.02) were noticed between age and year and vice versa. These correlations signify the impact that these features have upon one another.

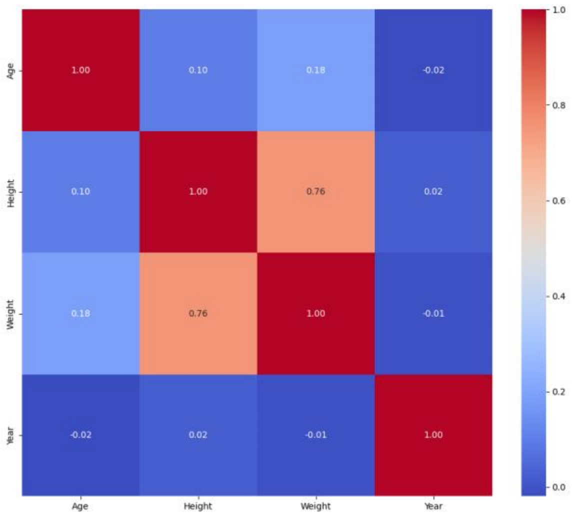

Fig. 2. Correlation heatmap [15].

Thirteen machine learning models, including logistic regression, polynomial logistic regression, XGBoost, Random Forest, KNN, Naive Bayes, LightGBM, AdaBoost, Decision Tree, Extra Trees, Gradient Boosting, Neural Network, and SVM, were selected for training and evaluation. The hyperparameters as shown in Table I were used to train these models. The chosen hyperparameters for each machine learning model were selected based on commonly recommended values in the literature and practical experience, covering key parameters such as learning_rate, n_estimators, and max_depth for ensemble models, C and gamma for SVM, and architecture-specific settings like the number of layers and neurons for Neural Networks, while systematic optimization methods such as GridSearchCV with cross-validation were employed to ensure robust and fair model performance evaluation. The training process involved multiple steps. At first, the data was split into training and testing sets with an 80-20 split. StandardScaler was used to normalize the features, and KNN Imputer handled any remaining missing values. The Synthetic Minority Over-sampling Technique (SMOTE) was applied to address class imbalance. Then each model was trained using the resampled training data. Specific hyperparameters were tuned using GridSearchCV or manual grid search to find the optimal settings for each algorithm. After that, the models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC values. Additionally, multiclass ROC curves were plotted for each model to visualize their performance in distinguishing between different medal types.

TABLE I. MODEL HYPERPARAMETERS

Model	Hyperparameters
Logistic Regression	random_state=42 max_iter=1000 class_weight=balanced
Polynomial Logistic Regression	degree = 4 random_state=42 max_iter=1000 class_weight=balanced
XGBoost	learning_rate= 0.001, 0.01, 0.1 n_estimators= 100,200,300,500 max_depth= 3, 6, 10 cv=5
Random Forest	n_estimators= 100, 200, 300 max_depth= 10, 15, 20, min_samples_split = 2, 5, 10 cv=5
KNN	n_neighbors= 5, 10, 20, 30 weights= uniform, distance_metric= euclidean, manhattan, chebyshev, minkowski cv=5
Naïve Nayes	nb = cuNB()
LightGBM	learning_rate = 0.01, 0.1, 0.3 n_estimators = 100, 200,300 max_depth= -1, 10, 20 num_leaves = 31, 50, 100 cv = 55
AdaBoost	n_estimators = 100, 200, 300 learning_rate = 0.001, 0.01, 0.1 cv = 5
Decision Tree	criterion = gini, entropy max_depth = 5, 10, 20, None min_samples_split = 2, 5, 10, 20 min_samples_leaf = 1, 2, 5 cv=5
Extra Tree	n_estimators = 100, 200 max_depth = 20, 30 min_samples_split = 5, 10 min_samples_leaf = 5, 10 max_features = sqrt, log2 cv=5
Gradient Boosting	learning_rate = 0.01, 0.1 n_estimators = 100, 200 max_depth = 3, 5, 10 cv =5
Neural Network	Loss function = Cross Entropy learning_rates = 0.001, 0.0005, 0.0001 early_stopping_patience = 5 num_epochs = 100 optimizer = Adam
SVM	kernel = rbf, probability = True C = 0.1, 1, 10, gamma = scale, auto cv = 5

The actual and predicted medal counts were compared for each model across the years 1896 to 2024. Line graphs illustrated the accuracy of the predictions of each model over time. The evaluation and comparison of these machine learning models highlighted their potential and limitations in forecasting future Olympic medal winners [5]-[10].

IV. RESULTS & DISCUSSION

The performance of the 13 machine learning models was evaluated using a range of metrics, including accuracy, precision, recall, F1-score, and ROC-AUC values. These metrics were calculated for each medal type (Gold, Silver, and Bronze), providing a comprehensive assessment of the predictive capabilities of each model.

Overall, from the three tables below (Table II, III, and IV), it is evident that the Ensemble models - XGBoost, LightGBM, and Gradient Boosting have performed quite better than the other models in the evaluation metrics. XGBoost, LightGBM, and Gradient Boosting had 83%, 84%, and 84% accuracies respectively, their precision was 0.11, 0.13, 0.09 for bronze, 0.13, 0.14, 0.14 for silver respectively. For gold all three of them had 25% precision.

TABLE II. EVALUATION METRICS FOR THE 13 MACHINE LEARNING MODELS FOR THE BRONZE MEDAL

Medal	Accuracy	Precision	Recall	F1-score	AUC value
Logistic Regression	48%	0.05	0.12	0.07	0.53
Polynomial Logistic Regression	47%	0.06	0.24	0.09	0.54
XGBoost	83%	0.11	0.04	0.05	0.60
Random Forest	77%	0.08	0.14	0.10	0.58
KNN	57%	0.07	0.19	0.10	0.53
Naïve Nayes	53%	0.06	0.13	0.08	0.55
LightGBM	84%	0.13	0.04	0.06	0.62
AdaBoost	72%	0.06	0.02	0.03	0.50
Decision Tree	74%	0.08	0.10	0.09	0.52
Extra Tree	74%	0.09	0.13	0.10	0.56
Gradient Boosting	84%	0.09	0.02	0.04	0.62
Neural Network	49%	0.06	0.21	0.09	0.55
SVM	47%	0.06	0.25	0.10	0.51

TABLE III. EVALUATION METRICS FOR THE 13 MACHINE LEARNING MODELS FOR THE SILVER MEDAL

Medal	Accuracy	Precision	Recall	F1-score	AUC value
Logistic Regression	48%	0.06	0.12	0.08	0.58
Polynomial Logistic Regression	47%	0.06	0.18	0.09	0.58
XGBoost	83%	0.13	0.05	0.07	0.63
Random Forest	77%	0.14	0.02	0.03	0.62
KNN	57%	0.07	0.19	0.10	0.53
Naïve Nayes	53%	0.07	0.11	0.08	0.57
LightGBM	84%	0.12	0.04	0.06	0.65
AdaBoost	72%	0.07	0.10	0.08	0.61
Decision Tree	74%	0.09	0.11	0.09	0.52
Extra Tree	74%	0.10	0.14	0.12	0.59
Gradient Boosting	84%	0.14	0.03	0.06	0.65
Neural Network	49%	0.06	0.22	0.10	0.57
SVM	47%	0.06	0.21	0.10	0.55

TABLE IV. EVALUATION METRICS FOR THE 13 MACHINE LEARNING MODELS FOR THE GOLD MEDAL

Medal	Accuracy	Precision	Recall	F1-score	AUC value
Logistic Regression	48%	0.07	0.41	0.12	0.60
Polynomial Logistic Regression	47%	0.09	0.32	0.13	0.63
XGBoost	83%	0.25	0.13	0.18	0.71
Random Forest	77%	0.18	0.26	0.21	0.69
KNN	57%	0.09	0.25	0.13	0.58
Naïve Nayes	53%	0.07	0.36	0.12	0.60
LightGBM	84%	0.25	0.13	0.17	0.73
AdaBoost	72%	0.11	0.26	0.15	0.64
Decision Tree	74%	0.13	0.18	0.15	0.56
Extra Tree	74%	0.15	0.23	0.18	0.68
Gradient Boosting	84%	0.25	0.12	0.16	0.73
Neural Network	49%	0.08	0.26	0.13	0.60
SVM	47%	0.08	0.28	0.12	0.61

XGBoost, LightGBM, and Gradient Boosting also had AUC values of 0.60, 0.62, 0.62 for bronze, 0.63, 0.65, 0.65 for silver, and 0.71, 0.73, and 0.73 for gold respectively, all of which are above 0.5, signifying that they should be able to give more stable predictions.

On the other hand, simpler models like Logistic Regression, Polynomial Logistic Regression, and SVM exhibited lower performances overall, while the other remaining models performed moderately.

The ROC curves for each model further highlighted their predictive capabilities. XGBoost and Gradient Boosting stood out with AUC values of 0.71 and 0.73 respectively as shown in Fig. 3, indicating strong discriminative ability. Ensemble methods like Random Forest (Fig. 4(a)) and Extra Trees also demonstrated high AUC values, reinforcing their robustness. In contrast, Logistic Regression, Polynomial Logistic Regression, KNN, and Decision Trees (Fig. 4(b)) had lower AUC values, reflecting their limited performance.

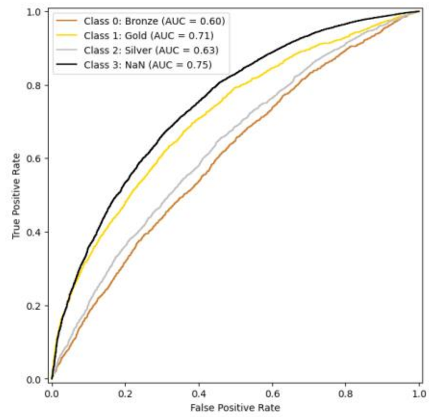

Fig. 3. ROC curves for XGBoost [15].

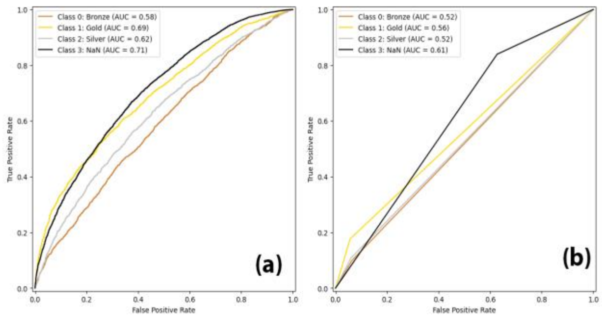

Fig. 4. ROC Curves for (a) Random Forest and (b) Decision Tree from 1896-2024 [15].

Although ensemble models such as LightGBM, XGBoost, and Gradient Boosting were supposed to demonstrate better alignment with the actual results according to the evaluation metrics, their prediction graphs below in Fig. 5(a) and Fig. 5(b) showed quite a significant difference between the actual and predicted results.

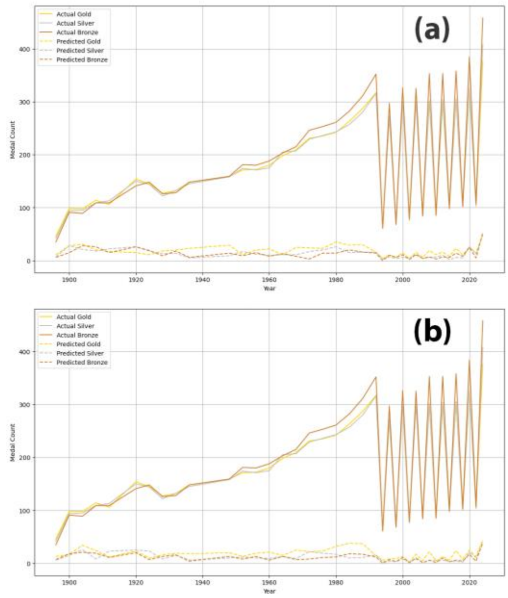

Fig. 5. Actual vs. Predicted Medal counts from 1869-2024 by (a) XGBoost and (b) LightGBM [15].

On the contrary, models like Logistic Regression, Polynomial Logistic Regression, KNN, and Naïve Bayes which showed lower results in their evaluation metrics, gave the above graphs in Fig. 6(a) and Fig. 6(b) which the predicted results were relatively closer to the actual results. This suggests that there might have either been some error in the training or the dataset in general for the prediction part.

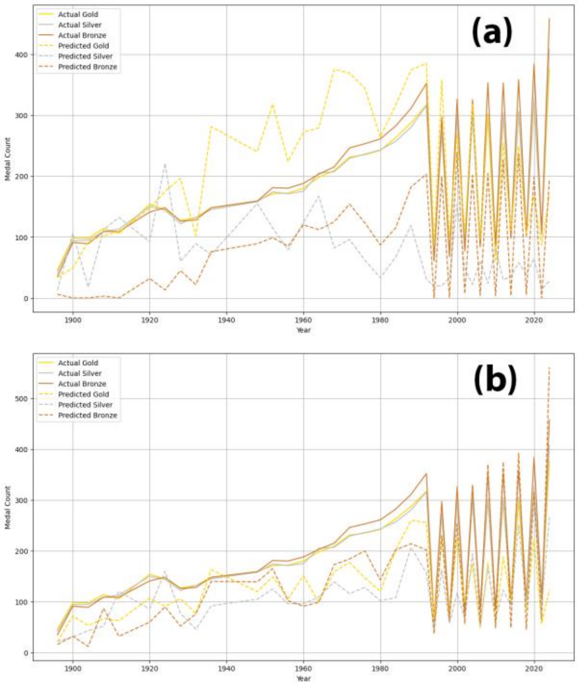

Fig. 6. Actual vs. Predicted Medal counts from 1869-2024 by (a) Logistic Regression and (b) Polynomial Logistic Regression [15].

And, the Neural Network showed the prediction values which were closest to the actual results of the Olympics from 1869-2024 as shown in Fig. 7.

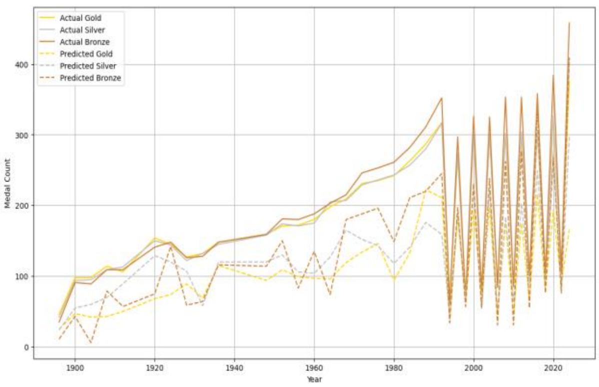

Fig. 7. Actual vs. Predicted Medal counts from 1869-2024 by Neural Network [15].

The results of the evaluation metrics indicated that ensemble models such as LightGBM, XGBoost, and Gradient Boosting outperformed other models in predicting Olympic medal outcomes. These models demonstrated higher accuracy and better AUC values, which should have shown better prediction when compared with actual results. The findings were consistent with previous studies that highlighted the effectiveness of boosting algorithms in predicting complex outcomes [1], [2]. The performance of simpler models such as Logistic Regression, Polynomial Logistic Regression, SVM, KNN, Naïve Bayes, and Decision Tree was significantly lower, indicating their limitations in handling this dataset. The other models showed moderate performance, suggesting the need for further tuning and optimization. However, their prediction graphs showed the opposite results. The ensemble models that performed better in the evaluation metrics illustrated predictions that had a significant gap from the actual medal results [16]. On the other hand, the poorly performing models displayed better prediction graphs. This indicates there might have been some issues in training the models or the dataset in general. Some discrepancies were also seen in prediction graphs for all the models somewhere around before 1990 [17]. Overall, the study provided valuable insights into the predictive capabilities of various machine learning models for Olympic medal outcomes. The findings could aid national Olympic committees in strategic planning and resource allocation for future Olympic Games.

V. Conclusion

In this study, various machine learning models were evaluated for their predictive capabilities regarding Olympic medal outcomes. A comprehensive dataset covering 128 years of Olympic history, from 1896 to 2024, was utilized to train and test thirteen machine learning models, including Logistic Regression, Polynomial Logistic regression, XGBoost, Random Forest, KNN, Naive Bayes, LightGBM, AdaBoost, Decision Tree, Extra Trees, Gradient Boosting, Neural Network, and SVM. It was observed that ensemble models such as XGBoost, LightGBM, and Gradient Boosting demonstrated superior performance metrics, with accuracy rates of 83%, 84%, and 84% respectively, and notable AUC values of 0.60, 0.62, and 0.62 for bronze medals, 0.63, 0.65, and 0.65 for silver medals, and 0.71, 0.73, and 0.73 for gold medals, reflecting their strong predictive capabilities. Conversely, simpler models like Logistic Regression, Polynomial Logistic Regression, and SVM exhibited lower performances overall. However, discrepancies in the prediction graphs indicated potential issues in the training or encoding process, suggesting that while evaluation metrics favored ensemble models, simpler models provided more stable and closer predictions in some cases, and Neural Network gave the closest predictions. This suggested potential errors in training or in the dataset in general for prediction.

In future research, it will be essential to explore the integration of additional features and more sophisticated preprocessing techniques to enhance model performance. The inclusion of socio-economic and geopolitical factors will be considered to provide a more comprehensive analysis of the factors influencing Olympic success [6], [7]. Additionally, advanced machine learning techniques such as deep learning should be investigated to further improve prediction accuracy. Furthermore, real-time data analysis and prediction could be implemented to provide timely insights for stakeholders in the sports industry. The development of interactive dashboards and visualization tools will be prioritized to facilitate the interpretation and application of the predictive results of the models.

Overall, the findings of this study will contribute to the field of sports analytics and aid national Olympic committees in strategic planning and resource allocation for future Olympic Games. The continuous improvement and validation of predictive models will be crucial in providing accurate and actionable insights for enhancing athletic performance and achieving Olympic success.

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